2. Tentative Schedule

Subject to change.

Week	Week of	Topics
1	Aug. 31	Foundations of Knowledge Discovery & the Data Mining Process; Python Basics
2	Sept. 7	Core Concepts in Data, Statistics, & Computational Thinking; Descriptive Statistics & Data Visualization
3	Sept. 14	Core Concepts in Data, Statistics, & Computational Thinking (cont'd); Exploratory Data Analysis (EDA) I
4	Sept. 21	Data Preparation, Similarity & Distance Measures, & Feature Selection; EDA II & Case Studies
5	Sept. 28	Dimensionality Reduction; K-Nearest Neighbors
6	Oct. 5	Data Mining Approaches: Predictive Modeling – Regression
7	Oct. 12	Data Mining Approaches: Predictive Modeling – Classification I
8	Oct. 19	Data Mining Approaches: Predictive Modeling – Classification II & Model Evaluation
9	Oct. 26	*Fall Break*
10	Nov. 2	Data Mining Approaches: Clustering
11	Nov. 9	Data Mining Approaches: Association Analysis & Anomaly Detection
12	Nov. 16	Working with Databases & Large Datasets: Relational & Nonrelational Query Formulation, incl. Biomedical & Health Databases
13	Nov. 23	Evaluation, Validation, & Interpretation of Results; Sequence Analysis & Modeling (Bioinformatics: BLAST, Hidden Markov Models)
14	Nov. 30	Contemporary Topics & Trends in Data Mining: Deep Learning & Neural Networks
15	Dec. 7	Applied Knowledge Discovery Across Domains (Health, Bioinformatics, Business); Final Project Presentations & Course Wrap-Up
16	Dec. 14	Final Examination & Evaluation Week